

TEMPERATURE SENSOR

H Series | External Probe

Wireless Temperature Monitoring



KEY INDICATORS



Measuring Range

-200°C to +800°C*



Accuracy

±0.5°C
±0.1°C option*



Wireless

4G / NB-IoT
/ LoRa



Protection

IP67/IP68
Transmitter enclosure



Battery Life

Over 10 years*
10-min sampling
Hourly reporting

OVERVIEW

The Hitelecom H Series wireless temperature sensor brings readings from distributed measurement points to Hitelecom Cloud or a compatible customer platform. Select the probe, enclosure and 4G, NB-IoT or LoRa variant for the installation. Suitable configurations monitor air, surfaces or liquids in industrial facilities, cold chains, laboratories, agriculture and temperature-sensitive storage.

FEATURES

CLOUD

Connect to Hitelecom Cloud or a compatible customer platform for temperature data and trends.

GREEN

Low-power design: over 10 years with 10-min sampling and 1-hour reporting at 25°C with normal signal.*

IP68

IP68-rated transmitter enclosure options for demanding indoor and outdoor installations.

NFC

NFC activation and local configuration on supported models; select the method with your variant.

REMOTE

Remote readings and configurable threshold alerts help teams respond to temperature changes.

CUSTOMIZATION

Match probe, cable, working range, enclosure and wireless connection to the installation.



Representative application environments; sensing assemblies are selected for each installation.

Item		Description
Physical Characteristics	Product models	H200 / H300 / H500
	Power supply	Battery-powered; battery pack selected by configuration.
	Battery life	Over 10 years: 10-min sampling, 1-hour reporting; 25°C ambient, normal signal.*
	SIM card	Type1: Standard IoT SIM Card Type2: Embedded SIM (eSIM option)
	Antenna	Built-in
	Size	135*66*66
	Data protocol	MQTT; agree the data path for the selected wireless variant.
	Reporting profile	Specify sampling, reporting and alert requirements.
	Local configuration	NFC activation and local setup, where fitted.
	Mounting	Mounting lugs / pole clamp / slotted mount.
Connectivity	Wireless options	4G / NB-IoT / LoRa, by selected variant.
	LTE-FDD bands	B1 / B3 / B5 / B8
	LTE-TDD bands	B34 / B38 / B39 / B40 / B41
Temperature Measurement	Measuring range	Family envelope: -200°C to +800°C; selected probe / input.*
	Accuracy	±0.5°C; ±0.1°C option for an agreed range and configuration.*
	Probe connection	Three-wire connection listed in the family reference.
	Measurement mode	Air / surface / immersion, with a suitable sensing assembly.
	Probe and cable	Probe type, probe length and cable length can be customized.

LTE bands shown are the published family reference. Regional modem, SIM and radio settings depend on the supplied variant.

	Item	Description
Waterproof Rating	Physical body protection	IP67/IP68 (transmitter enclosure)
	Ingress protection	IP68
Reliability	Electrostatic discharge immunity	Contact discharge: 6 kV; air discharge: 8 kV
	Surge immunity	Differential mode: 1 kV; common mode: 2 kV
	Fast transient burst	Power supply circuit: 2 kV
Environmental Range	Operating temperature	-30°C to +75°C (transmitter platform reference)
	Operating humidity	5% to 95% RH (non-condensing)
	Storage temperature	-40°C to +85°C (transmitter platform reference)
Integration	Data destination	Hitelecom Cloud or a compatible customer platform.
	LoRa integration	Match the radio protocol and compatible receiver / gateway.
	Threshold alerts	Agree temperature limits and the notification workflow.
Certification References CCC / CE / FCC / RoHS	Safety	EN IEC 62368-1:2020+A11:2020
	Health	EN IEC 62311:2020
	EMC	ETSI EN 301 489-1 V2.2.3 (2019-11) ETSI EN 301 489-52 V1.2.1 (2021-11)
	Radio	ETSI EN 301 908-1 V15.2.1 (2023-01) ETSI EN 301 908-13 V13.2.1 (2022-02)
	Frequency range	FDD B1: 1920-1980 MHz TX; 2110-2170 MHz RX FDD B3: 1710-1785 MHz TX; 1805-1880 MHz RX FDD B8: 880-915 MHz TX; 925-960 MHz RX TDD B40: 2300-2400 MHz TX / RX
	Maximum output power	FDD B1: 22.89 dBm max. FDD B3: 23.23 dBm max. FDD B8: 24.02 dBm max. TDD B40: 22.24 dBm max.
	Modulation	QPSK, 16QAM

* Range / accuracy: select the probe and input for the working range. Probe measurement range and transmitter ambient operating range are separate specifications.

* Battery life: over 10 years with sampling every 10 minutes and reporting every 1 hour, at 25°C ambient with normal signal. Actual service life depends on battery, network and operating conditions.

Platform references: reliability, environmental limits and compliance / radio values are retained from the supplied datasheets; confirm applicability and certificates for the supplied H Series temperature configuration.



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